

CLASH OF THE **CLAWS**

R V College of Engineering

The Devil's Advocate

The AI companion for balanced research

Samitha S B

Gauri G D

Mohit D K

Dia Arora



Problem Statement

Modern online research is often unintentionally biased.

- **Personalized algorithms** keep showing users content aligned with their existing views
- Studies show **that confirmation bias** makes people naturally seek evidence that supports what they already believe
- AI tools summarize information quickly, but users **rarely verify source diversity**
- Information overload pushes users toward **faster answers instead of balanced exploration**

Our target audience and Importance:

it is most useful for:

- **Researchers:** Push them to understanding all opposing claims in existing research
- **Students:** Students researching for essays/papers/ general knowledge may not encounter many strong opposing arguments which diversifies their thinking

However, this tool **can be used by anyone** allowing for balanced research on a topic, enabling them to understand additional perspectives during a research session in real time. Pushing for more holistic research as such ensures well rounded **awareness** on topics and **prevents bias from partial truths**

Why we chose the theme productivity:

Our product focuses on improving the quality and efficiency of research. Modern research is a productivity task—students, professionals, analysts, and teams spend significant time gathering information, validating sources, and making decisions.

Biased or one-sided research leads to wasted time, poor decisions, and repeated effort to manually verify alternative perspectives.

Current Solution & Gaps

Existing Solutions

Today, users commonly rely on tools such as [ChatGPT](#), [Google Gemini](#), [Google Scholar](#), and [Semantic Scholar](#) for research assistance and quick information summarization. These platforms improve search speed, generate summaries, and help users discover relevant papers or articles.

Gaps in Existing Systems

However, these tools still have major limitations:

- Most systems optimize for relevance and personalization, which can unintentionally reinforce confirmation bias.
- AI chatbots may generate misleading or one-sided responses without actively encouraging opposing viewpoints.
- Academic search engines like Google Scholar and Semantic Scholar can also show biased search patterns depending on how queries are phrased.
- Existing tools focus mainly on summarizing information faster rather than helping users explore balanced perspectives.

Opportunity

There is currently no mainstream research assistant that continuously monitors a user's research journey and actively introduces credible counter perspectives in real time. This creates an opportunity for a system focused on balanced and holistic research rather than just faster answers.

Your Solutions

What We Built

A Chrome extension + AI backend that analyzes a user's research session and introduces balanced counter perspectives.

Core Features

1. Session-Aware Bias Auditor

Tracks articles visited during a session and estimates overall bias score.

2. Counter Perspective Generator

Finds opposing viewpoints with supporting sources.

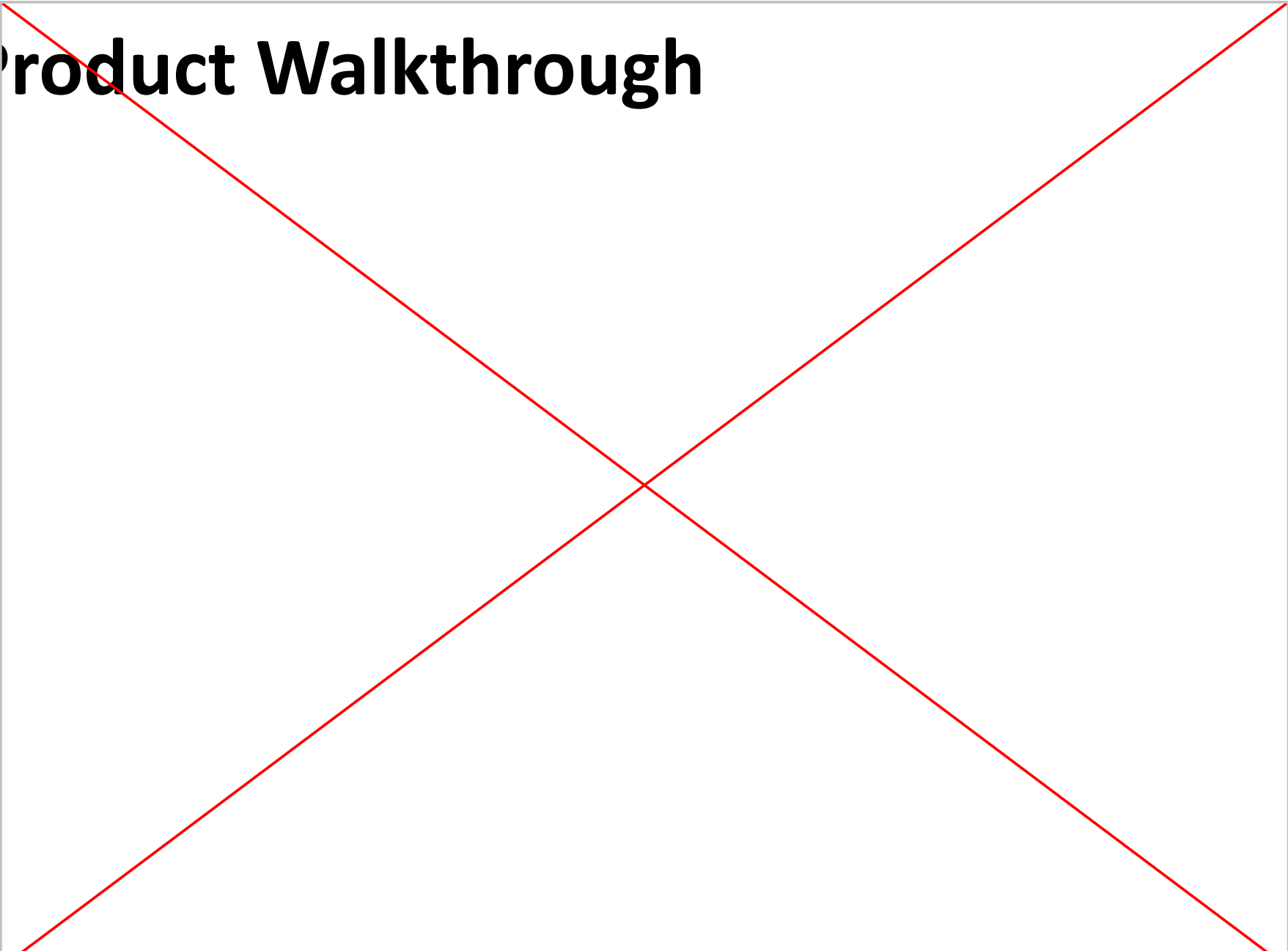
3. 4-Level Safety Gate

Filters irrelevant, unsafe, or low-quality pages before analysis.

4. Real-Time Research Companion

Works while users browse normally.

Demo/Product Walkthrough



Tech & Architecture

Tech Stack

Frontend

- Chrome Extension
- JavaScript
- HTML/CSS

Backend

- FastAPI
- Python

AI + APIs

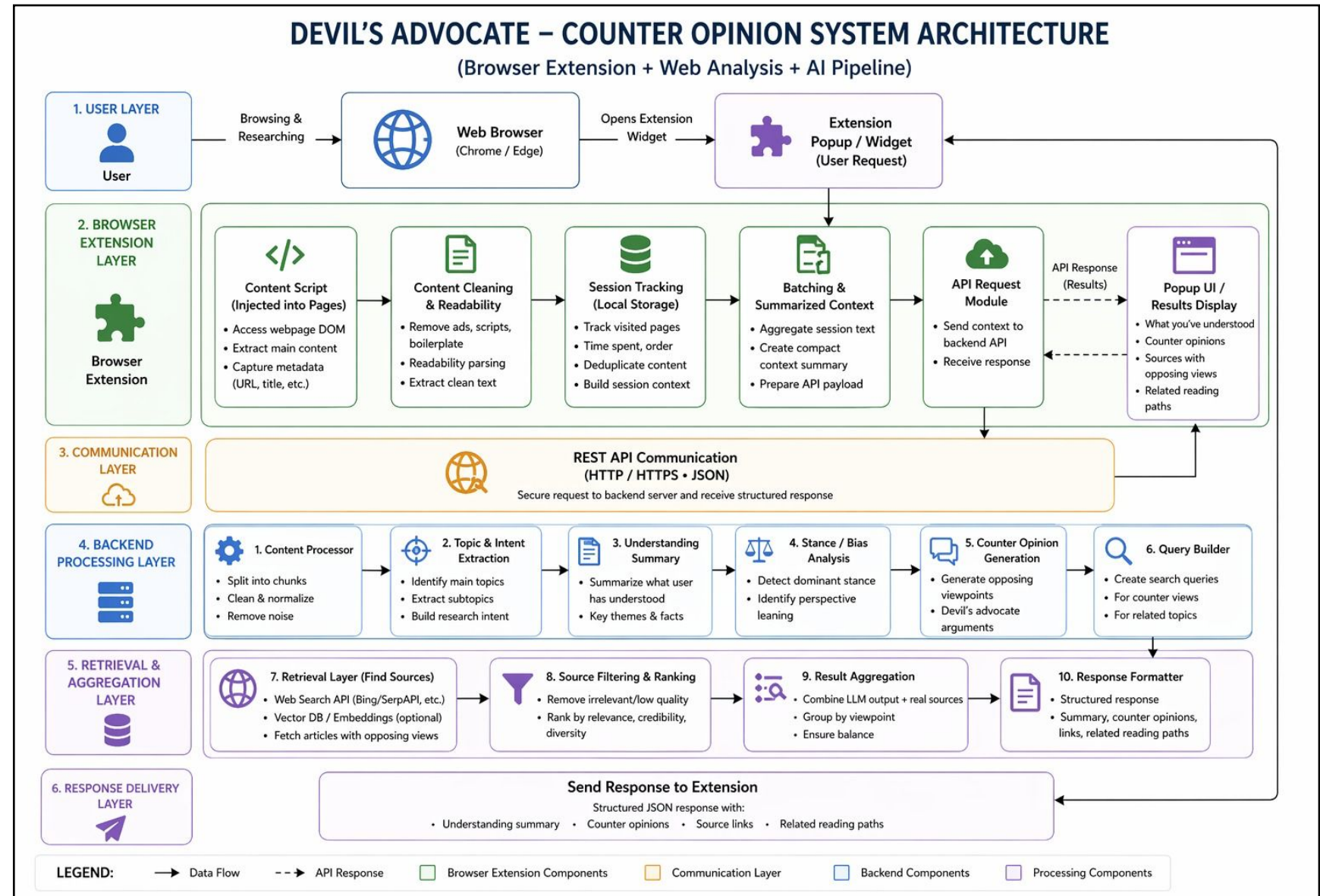
- **Groq API** (LLM inference)
- **SerpAPI** (source retrieval)
- **Pinecone** (vector similarity)

Tech & Architecture

AI Agents

- Gatekeeper Agent
- Bias Auditor
- Devil's Advocate Agent
- Librarian Agent

Workflow



Impact & Use cases

Students

Helps explore multiple perspectives while writing essays or projects.

Researchers

Encourages balanced literature review and critical analysis.

Journalists / Analysts

Prevents one-sided information gathering.

General Users

Improves awareness and reduces echo chambers online.

Scale Potential

Can expand into:

- education platforms
- enterprise research tools
- newsroom workflows
- browser productivity suites

Differentiation

Why Our Solution Stands Out

Most AI research tools focus on summarizing information quickly, but they do not actively address confirmation bias. As a result, users are often exposed to repetitive viewpoints that reinforce their existing opinions.

The Devil's Advocate takes a different approach by encouraging balanced research in real time. Instead of analyzing only a single article, the system studies the user's overall research session to identify patterns of bias and missing perspectives.

Another key differentiator is our hallucination-aware design. The system avoids generating unsupported counterarguments and prioritizes credible, source-backed perspectives, making the research process more trustworthy and holistic.

Demo Link

GitHub Repo- <https://github.com/gauri-dhanakshirur/The-Devils-Advocate.git>

Demo Link-

https://drive.google.com/file/d/1Q4UrW-SWcECwRRUYOdWiqDpWEAQSe-ay/view?usp=drive_link